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Analyzing Income Statements

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EARNINGS PER SHARE

Learning Outcome

- describe how earnings per share is calculated and calculate and interpret a company's basic and diluted earnings per share for companies with simple and complex capital structures including those with antidilutive securities

One income statement metric of particular importance to equity investors is earnings per share (EPS). IFRS require the presentation of EPS on the face of the income statement for net profit or loss (net income) and profit or loss (income) from continuing operations and similar presentation is required under US GAAP. This lesson outlines the calculations for EPS and explains how the calculation differs for a simple versus complex capital structure.

Simple versus Complex Capital Structure

A company's capital is composed of its equity and debt. Some types of equity have preference over others, and some debt (and other instruments) may be converted into equity. Under IFRS, the type of equity for which EPS is presented is referred to as ordinary. **Ordinary shares** are those equity shares that are subordinate to all other types of equity. The ordinary shareholders are basically the owners of the company—the equity holders who are paid last in a liquidation of the company and who benefit the most when the company does well. Under US GAAP, this ordinary equity is referred to as **common stock** or **common shares**, reflecting US language usage. The terms "ordinary shares," "common stock," and "common shares" are used interchangeably in the following discussion.

When a company has issued any financial instruments that are potentially convertible into common stock, it is said to have a complex capital structure. Examples of financial instruments that are potentially convertible into common stock include convertible bonds, convertible preferred stock, employee stock options, and warrants (a warrant is essentially an equity call option issued by the company; a warrant holder has the right but not the obligation to purchase newly issued shares at the exercise price). If a company's capital structure does not include such potentially convertible financial instruments, it is said to have a simple capital structure.

The distinction between simple versus complex capital structure is relevant to the calculation of EPS because financial instruments that are potentially convertible into common stock could, as a result of conversion or exercise, potentially dilute (i.e., decrease) EPS. Information about such a potential dilution is valuable to a company's current and potential shareholders; therefore, accounting standards require companies to disclose what their EPS would be if all dilutive financial instruments were converted into common stock. The EPS that would result if all dilutive financial instruments were converted is called **diluted EPS**. In contrast, **basic EPS** is calculated using the reported earnings available to common shareholders of the parent company and the weighted average number of shares outstanding.

Companies are required to report both basic and diluted EPS as well as amounts for continuing operations. Exhibit 12 shows the per share amounts reported by AB InBev at the bottom of its income statement. The company's basic EPS ("before dilution") was USD4.06, and diluted EPS ("after dilution") was USD3.98 for 2017. In addition, in the same way that AB InBev's income statement shows income from continuing operations separately from total income, EPS from continuing operations is also shown separately from total EPS. For 2017, the basic and diluted EPS from continuing operations were USD4.04 and USD3.96, respectively. Across all measures, AB InBev's EPS was much

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Exhibit 12: AB InBev's Earnings per Share (USD)

	12 Months Ended 31 December		
	2017	2016	2015
Basic earnings per share	4.06	0.72	5.05
Diluted earnings per share	3.98	0.71	4.96
Basic earnings per share from continuing operations	4.04	0.69	5.05
Diluted earnings per share from continuing operations	3.96	0.68	4.96

Basic EPS

Basic EPS is the amount of income available to common shareholders divided by the weighted average number of common shares outstanding over a period. The amount of income available to common shareholders is the amount of net income remaining after preferred dividends (if any) have been paid. Thus, the formula to calculate basic EPS is as follows:

$$\text{Basic EPS} = \frac{\text{Net income} - \text{Preferred dividends}}{\text{Weighted average number of shares outstanding}}.$$

The weighted average number of shares outstanding is a time weighting of common shares outstanding. For example, assume a company began the year with 2,000,000 common shares outstanding and repurchased 100,000 common shares on 1 July. The weighted average number of common shares outstanding would be the sum of 2,000,000 shares × 1/2 year + 1,900,000 shares × 1/2 year, or 1,950,000 shares. So, the company would use 1,950,000 shares as the weighted average number of shares in calculating its basic EPS.

If the number of shares of common stock increases as a result of a stock dividend or a stock split, the EPS calculation reflects the change retroactively to the beginning of the period.

Example 8, 9, and 10 illustrate the computation of basic EPS.

EXAMPLE 8

A Basic EPS Calculation (1)

1. For the year ended 31 December 2018, Shopalot Company had net income of USD1,950,000. The company had 1,500,000 shares of common stock outstanding, no preferred stock, and no convertible financial instruments. What is Shopalot's basic EPS?

   Hide Solution

   Solution:

   Shopalot's basic EPS is USD1.30 (USD1,950,000 divided by 1,500,000 shares).

EXAMPLE 9

A Basic EPS Calculation (2)

For the year ended 31 December 2018, Angler Products had net income of USD2,500,000. The company declared and paid USD200,000 of dividends on

Exhibit 13: Angler's Common Stock Shares

Shares outstanding on 1 January 2018	1,000,000
Shares issued on 1 April 2018	200,000
Shares repurchased (treasury shares) on 1 October 2018	(100,000)
Shares outstanding on 31 December 2018	1,100,000

1. What is the company's weighted average number of shares outstanding?

Hide Solution

Solution:

The weighted average number of shares outstanding is determined by the length of time each quantity of shares was outstanding:

1,000,000 × (3 months/12 months) =	250,000
1,200,000 × (6 months/12 months) =	600,000
1,100,000 × (3 months/12 months) =	275,000
Weighted average number of shares outstanding	1,125,000

2. What is the company's basic EPS?

Hide Solution

Solution:

Basic EPS = (Net income – Preferred dividends)/Weighted average number of shares = (USD2,500,000 – USD200,000)/1,125,000 = USD2.04

EXAMPLE 10

A Basic EPS Calculation (3)

1. Assume the same facts as Example 9 except that on 1 December 2018, a previously declared 2-for-1 stock split took effect. Each shareholder of record receives two shares in exchange for each current share that he or she owns. What is the company's basic EPS?

Hide Solution

Solution:

For EPS calculation purposes, a stock split is treated as if it occurred at the beginning of the period. The weighted average number of shares would, therefore, be 2,250,000, and the basic EPS would be USD1.02 [= (USD2,500,000 – USD200,000)/2,250,000].

Diluted EPS: The If-Converted Method

If a company has a simple capital structure (in other words, one that includes no potentially dilutive financial instruments), then its basic EPS is equal to its diluted EPS. If, however, a company has potentially dilutive financial instruments, its diluted EPS may differ from its basic EPS. Diluted EPS, by definition, is always equal to or less than basic EPS. The following sections describe the effects of three types of potentially dilutive financial instruments on diluted EPS: convertible preferred, convertible debt, and

Outstanding

When a company has convertible preferred stock outstanding, diluted EPS is calculated using the **if-converted method**. The if-converted method is based on what EPS would have been if the convertible preferred securities had been converted at the beginning of the period. In other words, the method calculates what the effect would have been if the convertible preferred shares converted at the beginning of the period. If the convertible shares had been converted, there would be two effects. First, the convertible preferred securities would no longer be outstanding; instead, additional common stock would be outstanding. Thus, under the if-converted method, the weighted average number of shares outstanding would be higher than in the basic EPS calculation. Second, if such a conversion had taken place, the company would not have paid preferred dividends. Thus, under the if-converted method, the net income available to common shareholders would be higher than in the basic EPS calculation.

Diluted EPS using the if-converted method for convertible preferred stock is equal to net income divided by the weighted average number of shares outstanding from the basic EPS calculation plus the additional shares of common stock that would be issued upon conversion of the preferred. Thus, the formula to calculate diluted EPS using the if-converted method for preferred stock is as follows:

$$\text{Diluted EPS} = \frac{\text{(Net income)}}{\text{(Weighted average number of shares outstanding + New common shares that would have been issued at conversion)}}.$$

A diluted EPS calculation using the if-converted method for preferred stock is provided in Example 11.

EXAMPLE 11

A Diluted EPS Calculation Using the If-Converted Method for Preferred Stock

1. For the year ended 31 December 2018, Bright-Warm Utility Company (fictitious) had net income of USD1,750,000. The company had an average of 500,000 shares of common stock outstanding, 20,000 shares of convertible preferred, and no other potentially dilutive securities. Each share of preferred pays a dividend of USD10 per share, and each is convertible into five shares of the company's common stock. Calculate the company's basic and diluted EPS.

Hide Solution

Solution:

If the 20,000 shares of convertible preferred had each converted into five shares of the company's common stock, the company would have had an additional 100,000 shares of common stock (five shares of common for each of the 20,000 shares of preferred). If the conversion had taken place, the company would not have paid preferred dividends of USD200,000 (USD10 per share for each of the 20,000 shares of preferred). As shown in Exhibit 14, the company's basic EPS was USD3.10 and its diluted EPS was USD2.92.

Exhibit 14:

Calculation of Diluted EPS for Bright-Warm Utility Company Using the If-Converted Method: Case of Preferred Stock

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Preferred dividend	–200,000	0
Numerator	USD1,550,000	USD1,750,000
Weighted average number of shares outstanding	500,000	500,000
Additional shares issued if preferred converted	0	100,000
Denominator	500,000	600,000
EPS	**USD3.10**	**USD2.92**

Diluted EPS When a Company Has Convertible Debt Outstanding

When a company has convertible debt outstanding, the diluted EPS calculation also uses the if-converted method. Diluted EPS is calculated as if the convertible debt had been converted at the beginning of the period. If the convertible debt had been converted, the debt securities would no longer be outstanding; instead, additional shares of common stock would be outstanding. Also, if such a conversion had taken place, the company would not have paid interest on the convertible debt, so the net income available to common shareholders would increase by the after-tax amount of interest expense on the debt converted.

Thus, the formula to calculate diluted EPS using the if-converted method for convertible debt is as follows:

$$\text{Diluted EPS} = \frac{(\text{Net income} + \text{After-tax interest on convertible debt} - \text{Preferred dividends})}{\begin{array}{c}(\text{Weighted average number of shares outstanding} + \text{Additional} \\ \text{common shares that would have been issued at conversion})\end{array}}.$$

A diluted EPS calculation using the if-converted method for convertible debt is provided in Example 12.

EXAMPLE 12

A Diluted EPS Calculation Using the If-Converted Method for Convertible Debt

1. Oppnox Company (fictitious) reported net income of USD750,000 for the year ended 31 December 2018. The company had a weighted average of 690,000 shares of common stock outstanding. In addition, the company has only one potentially dilutive security: USD50,000 of 6 percent convertible bonds, convertible into a total of 10,000 shares. Assuming a tax rate of 30 percent, calculate Oppnox's basic and diluted EPS.

   Hide Solution

   Solution:

   If the debt securities had been converted, the debt securities would no longer be outstanding and instead, an additional 10,000 shares of common stock would be outstanding. Also, if the debt securities had been converted, the company would not have paid interest of USD3,000 on the debt, so net income available to common shareholders would have increased by USD2,100 [= USD3,000(1 – 0.30)] on an after-tax basis. Exhibit 15 illustrates the calculation of diluted EPS using the if-converted method for convertible debt.

Exhibit 15:

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	Basic EPS	Diluted EPS Using If-Converted Method
Net income	USD750,000	USD750,000
After-tax cost of interest		2,100
Numerator	USD750,000	USD752,100
Weighted average number of shares outstanding	690,000	690,000
If converted	0	10,000
Denominator	690,000	700,000
EPS	**USD1.09**	**USD1.07**

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Diluted EPS: The Treasury Stock Method

When a company has stock options, warrants, or their equivalents outstanding, diluted EPS is calculated as if the financial instruments had been exercised and the company had used the proceeds from exercise to repurchase as many shares of common stock as possible at the average market price of common stock during the period. The weighted average number of shares outstanding for diluted EPS is thus increased by the number of shares that would be issued upon exercise minus the number of shares that would have been purchased with the proceeds. This method is called the treasury stock method under US GAAP because companies typically hold repurchased shares as treasury stock. The same method is used under IFRS but is not named.

For the calculation of diluted EPS using this method, the assumed exercise of these financial instruments would have the following effects:

- The company is assumed to receive cash upon exercise and, in exchange, to issue shares.
- The company is assumed to use the cash proceeds to repurchase shares at the weighted average market price during the period.

As a result of these two effects, the number of shares outstanding would increase by the incremental number of shares issued (the difference between the number of shares issued to the holders and the number of shares assumed to be repurchased by the company). For calculating diluted EPS, the incremental number of shares is weighted based upon the length of time the financial instrument was outstanding in the year. If the financial instrument was issued before the beginning of the year, the weighted average number of shares outstanding increases by the incremental number of shares. If the financial instruments were issued during the year, then the incremental shares are weighted by the amount of time the financial instruments were outstanding during the year.

The assumed exercise of these financial instruments would not affect net income. For calculating EPS, therefore, no change is made to the numerator. The formula to calculate diluted EPS using the treasury stock method (same method as used under IFRS but not named) for options is as follows:

$$\text{Diluted EPS} = \frac{\text{(Net income - Preferred dividends)}}{\begin{array}{c}\text{[Weighted average number of shares} \\ \text{outstanding + (New shares that would} \\ \text{have been issued at option exercise -} \\ \text{Shares that could have been purchased} \\ \text{with cash received upon exercise) } \times \\ \text{(Proportion of year during which the} \\ \text{financial instruments were outstanding)]}\end{array}}$$

EXAMPLE 13

A Diluted EPS Calculation Using the Treasury Stock Method for Options

1. Hihotech Company (fictitious) reported net income of USD2.3 million for the year ended 30 June 2018 and had a weighted average of 800,000 common shares outstanding. At the beginning of the fiscal year, the company has outstanding 30,000 options with an exercise price of USD35. No other potentially dilutive financial instruments are outstanding. Over the fiscal year, the company's market price has averaged USD55 per share. Calculate the company's basic and diluted EPS.

Hide Solution

Solution:

Using the treasury stock method, we first calculate that the company would have received USD1,050,000 (USD35 for each of the 30,000 options exercised) if all the options had been exercised. The options would no longer be outstanding; instead, 30,000 shares of common stock would be outstanding. Under the treasury stock method, we assume that shares would be repurchased with the cash received upon exercise of the options. At an average market price of USD55 per share, the USD1,050,000 proceeds from option exercise, the company could have repurchased 19,091 shares. Therefore, the incremental number of shares issued is 10,909 (calculated as 30,000 minus 19,091). For the diluted EPS calculation, no change is made to the numerator. As shown in Exhibit 16, the company's basic EPS was USD2.88 and the diluted EPS was USD2.84.

Exhibit 16:

Calculation of Diluted EPS for Hihotech Company Using the Treasury Stock Method: Case of Stock Options

	Basic EPS	Diluted EPS Using Treasury Stock Method
Net income	USD2,300,000	USD2,300,000
Numerator	USD2,300,000	USD2,300,000
Weighted average number of shares outstanding	800,000	800,000
If converted	0	10,909
Denominator	800,000	810,909
EPS	**USD2.88**	**USD2.84**

As noted, IFRS require a similar computation but does not refer to it as the "treasury stock method." The company is required to consider that any assumed proceeds are received from the issuance of new shares at the average market price for the period. These new "inferred" shares would be disregarded in the computation of diluted EPS, but the excess of the new shares that would be issued under options contracts minus the new inferred shares would be added to the weighted average number of shares outstanding. The results are the same as the treasury stock method, as shown in Example 14.

1. Assuming the same facts given in Example 13, calculate the weighted average number of shares outstanding for diluted EPS under IFRS.

Hide Solution

Solution:

If the options had been exercised, the company would have received USD1,050,000. If this amount had been received from the issuance of new shares at the average market price of USD55 per share, the company would have issued 19,091 shares. IFRS refer to the 19,091 shares the company would have issued at market prices as the inferred shares. The number of shares issued under options (30,000) minus the number of inferred shares (19,091) equals 10,909. This amount is added to the weighted average number of shares outstanding of 800,000 to get diluted shares of 810,909. Note that this is the same result as that obtained under US GAAP; it is just derived in a different manner.

Other Issues with Diluted EPS and Changes in EPS

It is possible that some potentially convertible securities could be **antidilutive** (i.e., their inclusion in the computation would result in an EPS higher than the company's basic EPS). Under IFRS and US GAAP, antidilutive securities are not included in the calculation of diluted EPS. Diluted EPS should reflect the maximum potential dilution from conversion or exercise of potentially dilutive financial instruments. Diluted EPS will always be less than or equal to basic EPS. Example 15 provides an illustration of an antidilutive security.

EXAMPLE 15

An Antidilutive Security

1. For the year ended 31 December 2018, Dim-Cool Utility Company (fictitious) had net income of USD1,750,000. The company had an average of 500,000 shares of common stock outstanding, 20,000 shares of convertible preferred, and no other potentially dilutive securities. Each share of preferred pays a dividend of USD10 per share, and each is convertible into three shares of the company's common stock. What was the company's basic and diluted EPS?

Hide Solution

Solution:

If the 20,000 shares of convertible preferred had each converted into three shares of the company's common stock, the company would have had an additional 60,000 shares of common stock (three shares of common for each of the 20,000 shares of preferred). If the conversion had taken place, the company would not have paid preferred dividends of USD200,000 (USD10 per share for each of the 20,000 shares of preferred). The effect of using the if-converted method would be EPS of USD3.13, as shown in Exhibit 17. Because this is greater than the company's basic EPS of USD3.10, the securities are said to be antidilutive and the effect of their conversion would not be included in diluted EPS. Diluted EPS would be the same as basic EPS (i.e., USD3.10).

Exhibit 17: Calculation for an Antidilutive Security

	Basic EPS	Method	
Net income	USD1,750,000	USD1,750,000	
Preferred dividend	–200,000	0	
Numerator	USD1,550,000	USD1,750,000	
Weighted average number of shares outstanding	500,000	500,000	
If converted	0	60,000	
Denominator	500,000	560,000	
EPS	**USD3.10**	USD3.13	←Exceeds basic EPS; security is antidilutive and, therefore, **not** included. **Reported diluted EPS= USD3.10**.

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Changes in EPS

Having explained the calculations of both basic and diluted EPS, we return to an examination of changes in EPS. As noted in Exhibit 12, AB InBev's fully diluted EPS from continuing operations increased from USD0.68 in 2016 to USD3.96 in 2017. In general, an increase in EPS results from an increase in net income, a decrease in the number of shares outstanding, or a combination of both. In the notes to its financial statements (not shown), AB InBev discloses that the weighted average number of shares for both the basic and fully diluted calculations was greater in 2017 than in 2016. Thus, for AB InBev, the improvement in EPS from 2016 to 2017 was driven by an increase in net income. Changes in the numerator and denominator explain the changes in EPS arithmetically. To understand the business drivers of those changes requires further research. Lesson 5 presents analytical tools that an analyst can use to highlight areas for further examination.

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Discussion

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AL For example 9, why is 1,200,000 × (6 months/12 months) =600,000 for April? April is the 4th month and the shares existed for less than 6 months.

Created 3 months ago by Alejandra Lesso Mosqueda 2 replies | Last Activity: 3 months ago
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LT Example 10: Beginning of the period is 1st of January or 1st of December?

Created 4 months ago by Leon Tomski 2 replies | Last Activity: 3 months ago
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AF The fiscal start and end dates should be clearly outlined in every question. The CFA Institute should fix these to avoid unneccesary confusion. The "start of the year" is not outlined at all here, I guess we're simply supposed to assume it's Jan 1, however the question also doesn't clarify that this question follows the calendar year rather than any other sort of fiscal.

Screenshot%202025-09-22%20234024.png (84.7 KB)

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should see that that preferred shares are each converted into 5 shares. So that's why in Diluted EPS we multiply the 20,000 preferred shares by 5 and not by 10. However in the Basic EPS we do multiply thte 20,000 by 10 because it is the amount paid to preffered share holders, and the profit left is given to common shareholders. IT IS AN IMPORTANT POINT THAT ALL MUST FOCUS, Thats why I shared it with you all ppl.

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SG

Can anyone explain, how weighted average no. of shares are calculated in example 9

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AA

In Example 10 there were 10,00,000 share outstanding after 2 for 1 split total outstanding at the beginning of the year will become 20,00,000. So, 20,00,000*3/12 for first 3 months = 5,00,000. On 1st Apr,2025 2,00,000 new shares issued so (20,00,000+2,00,000)*6/12 = 11,00,000 for 6 months till Oct,25. On 1st Oct,2025 company repurchased 1,00,000 shares so (22,00,000-1,00,000)*3/12 = 5,25,000. Adding all the 3 gives 21,25,000 shares outstanding. How come they got 22,50,000 shares outstanding is my question ?

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